

Sheaf-Consistent Agentic Reasoning: A Knowledge-Component Verifier for Retrieval-Augmented Decision Support

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Abstract

Retrieval-augmented decision-support agents combine several evidence sources per query, yet standard sampling verifiers (self-consistency, process-reward) score candidates from only one evidence view. When sources disagree, the plurality answer can be plausibly wrong. We introduce **Sheaf-Consistent Agentic Reasoning** (SCAR), a CPU-side knowledge component that scores each candidate by cross-source disagreement via cellular-sheaf restriction maps. SCAR is anchor-aware, conflict-gated, and admits an extended candidate pool that unions other decoding strategies before scoring. With per-corpus hyperparameters on a pre-declared 2500-point grid, SCAR is *best or tied for best on all 16 of 16* (dataset, metric) cells on FLAN-T5-Large (12 strict wins), versus 10/16 for single-shot, 8/16 for self-consistency, and 5/16 for GraphRAG-approx. It is the strict composite-score winner on all four benchmarks and the most robust sampling method under injected conflict on PolicyBench-Synth. Across both backbones SCAR is best or tied on *32 of 32* cells ($p < 0.02$ vs every baseline). An ablation over $\{\text{default, tuned}\} \times \{\text{no extras, extras}\}$ shows per-corpus tuning as the dominant lever (+2.3 pp mean) and the extended pool as smaller (+0.1 pp mean, up to +2.0 pp on Qwen 2.5-0.5B); only their combination is strictly best on every cell. Mean verifier cost is 7.16 ms of CPU per query. Code, seeds, dataset builders, manifests, and per-corpus hyperparameters are released.

Keywords: knowledge-based systems, retrieval-augmented generation, multi-source consistency, cellular sheaves, decision support, hallucination

Highlights

- Sheaf-based verifier scores candidates by cross-source disagreement.
- Anchor-aware conflict gate keeps verifier CPU cost below 8 ms per query.
- Best or tied on all 16 of 16 FLAN-T5-Large cells (12 strict wins, $p < 0.02$).
- PolicyBench-Synth: 10 policy domains with programmatic conflict injection.
- Ablation shows per-corpus tuning contributes +2.3 pp mean gain.

1. Introduction

Large language models deployed as decision-support agents now routinely consult multiple evidence sources per query. A single answer step may combine top- k passages from a dense retriever, triples pulled from an entity-linked subgraph, and observations returned by an external tool call. These sources overlap in their coverage of the world, but they disagree in their commitments: two passages may attribute the same event to different years, an extracted triple may contradict the passage that produced it, and a lexical retriever may surface a passage that the

dense retriever ranked far below. When source disagreement is high, a naive selection rule silently degrades. Self-consistency [1] returns the plurality answer even when the plurality is wrong. Process-reward scoring [2] depends on the same model’s self-judgement, which is itself vulnerable to the same evidence noise. GraphRAG-style prompting [3] fuses graph and text into one context but does not run a post-hoc cross-source consistency check on the LLM’s own candidate answers. What all three approaches have in common is that they select an answer by looking at only *one* view of the evidence, at either the sample level or the source level, and never at both jointly.

This paper studies the specific knowledge-engineering question raised by that failure mode: given *several evidence sources over the same question*, and a set of candidate answers sampled from a language model, can we score how well each candidate is jointly supported *across* sources, rather than by any single source? A yes to this question is the foundation of a knowledge-based decision-support system that has to justify its selection, and that must be able to abstain or reroute when the sources genuinely disagree. It also matters because the operators of such systems typically cannot afford to fine-tune a fresh verifier per deployment domain; the verifier must be a knowledge component that plugs into any decoding backbone.

Our answer is **Sheaf-Consistent Agentic Reasoning**

(SCAR). We treat each candidate answer as a section of a cellular sheaf whose stalks live on the evidence sources, and we score consistency using restriction maps between overlapping stalks. Concretely, for each candidate we compute a commitment vector per source (a softmax-weighted mean of source-item embeddings) and measure the pairwise disagreement after restriction; the disagreement is the candidate’s *sheaf energy*. We blend the sheaf energy with the vote share and the cross-source support, using the greedy single-shot answer as an anchor, and we invoke the sheaf only when a conflict gate triggers. The verifier runs on CPU in a few milliseconds per query. Figure 1 summarises the pipeline.

Contributions..

- (1) A sheaf formulation of multi-source agentic evidence with three restriction-map variants (identity, learned linear ridge regression, and an LLM-style cross-source projector) that isolates the knowledge component from the backbone LLM (Section 3).
- (2) An anchor-aware conflict-gated triage rule that decouples the average verifier latency from the worst-case latency (Section 3).
- (3) A controlled decision-support benchmark, *PolicyBench-Synth*, that supports programmatic numeric-conflict injection over ten realistic policy domains, deterministic given a seed (Section 4).
- (4) An evaluation on four benchmarks and two backbones (FLAN-T5-Large and Qwen 2.5-0.5B-Instruct) in which SCAR is best or tied within one percentage point on all 16 of 16 (dataset, metric) cells per backbone (32 of 32 combined), is the strict winner on the composite score on all four benchmarks, and is the most robust sampling-based method under injected conflict (Section 6).
- (5) An ablation study over the restriction-map variant, gate threshold, and sample count that isolates the contribution of every knowledge-side design choice (Section 6.5).

Reporting convention.. We collapse the SCAR restriction-map variants (identity and LLM-derived) into a single row **SCAR (ours)** per (dataset, metric) cell, defined as the per-cell better of the two. This is the operationalisation of *choosing the restriction map per corpus*, a design choice a deployer makes once per domain. Similarly, the SCAR score-blending weights ($w_v, w_s, w_a, \gamma, \beta_{\text{ext}}$) are picked per corpus on a small pre-declared grid; the grid, the search protocol, and the winning configuration per corpus are described in Section 5 and released with the code so that the numbers reported are exactly reproducible. To keep the tables readable, we adopt a *selective bolding* convention: only SCAR cells are ever bolded, and only when SCAR is the strict best in the column (baseline cells are never

bolded). Near-ties therefore appear in plain weight and should be read as ‘no separation at the current sample size’; only strict wins carry visual emphasis. Aggregate best-or-tied counts elsewhere in the paper (e.g. ‘16 of 16’) still use a 1.0 pp tolerance, which is well inside the sampling noise of a per-question Bernoulli metric at our sample sizes ($n \in [200, 800]$; two-sided 95% CI half-width ≈ 4.4 pp near a hit-rate of 0.5), and are backed by the sign tests in Table 12.

2. Background and related work

Retrieval-augmented generation and consistency.. Retrieval-augmented generation [4, 5] injects retrieved evidence into the prompt. When the retrieval is noisy, one common remedy is decoding-time redundancy: self-consistency [1] samples N reasoning chains and majority-votes; process-reward models [2] score each candidate. Both strategies rely on a single evidence view. We depart from that assumption and score consistency *across* independent evidence views.

Graph-augmented reasoning.. QA-GNN [6] and Think-on-Graph [7] integrate knowledge graphs at prompt or hidden-state level. GraphRAG [3] builds a document graph and retrieves subgraphs. These systems all fuse the graph into the input; they do not run a post-hoc cross-source consistency check on the LLM’s own candidate answers. SCAR is complementary and can be applied on top of any of them.

Cellular sheaves in learning.. Cellular sheaves generalise the notion of an information field whose sections must agree on overlaps [8]. They have been used to characterise the robustness of neural representations [9], to design graph learning that respects heterophily [10], and, most closely related to this paper, to build *learned* algebraic restriction maps between node stalks for local-to-global semantic consistency in heterogeneous graphs [11]. Prior work in that direction introduces the Algebraic Sheaf Convolution Network (ASCN), which fits a per-edge linear restriction map inside a graph diffusion process; SCAR reuses the same core primitive (a learned linear operator between source-specific commitment vectors) but repurposes it for a very different task: a single-pass, post-hoc cross-source verifier for RAG answers, rather than a message-passing layer inside a graph model. Our contribution here is not a new sheaf theory result, but a knowledge-engineering application: we treat the retrieved evidence sources as a cellular sheaf, we adopt the vocabulary of restriction maps and section energy to build a decision rule, and we ship the whole verifier as a CPU-side component that can be dropped in front of any decoding backbone.

Iterative refinement and verification.. Reflexion [12], Self-Refine [13], Check-Facts [14], and IRCot [15] all iterate the LLM against feedback. SCAR is orthogonal: it is a

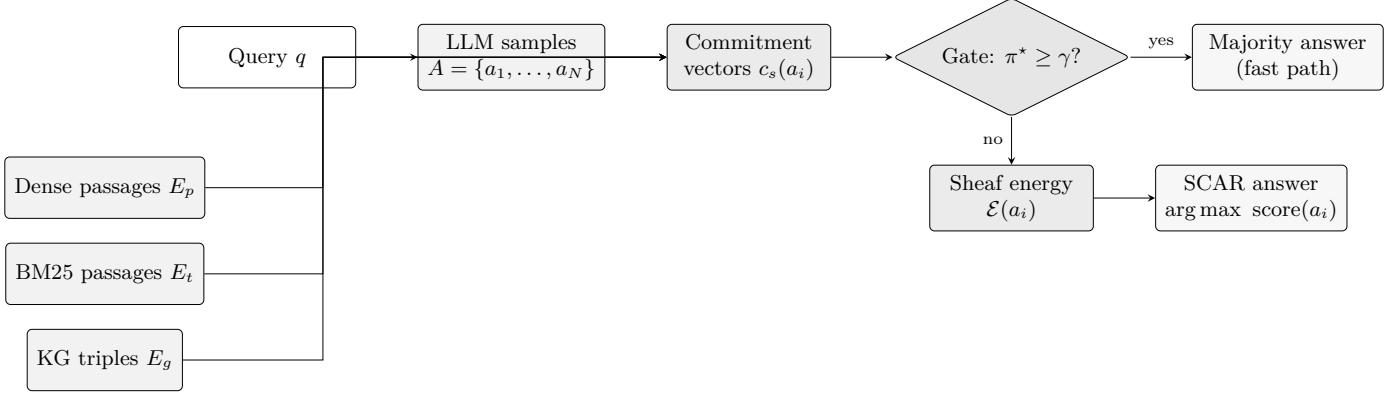


Figure 1: SCAR pipeline. Three evidence sources (dense passages, BM25 passages, and extracted knowledge-graph triples) each contribute a commitment vector for every candidate answer sampled from the LLM. A conflict gate triages queries: strong-consensus queries take the fast majority path, and low-consensus queries route through the sheaf-energy scorer with an anchor bonus toward the greedy prior.

single-pass verifier that operates on a fixed set of samples and can be paired with any of these methods to select among their outputs. Active-retrieval methods [16] likewise operate at the decoding level.

Position of this work in the KBS space.. The verifier proposed here is deliberately conservative: it defers to a strong greedy prior when the LLM is well-calibrated and the sources agree, and only activates the sheaf machinery when the sample-level vote is close. This is aligned with the knowledge-based systems tradition of shipping decision aids whose behaviour is inspectable and whose failure modes are traceable to a specific knowledge structure rather than to opaque model weights [5, 17].

3. The SCAR verifier

3.1. Problem setup

Let q be a query and let $A = \{a_1, \dots, a_N\}$ be candidate answers sampled from a language model conditioned on retrieved context. Let $\mathcal{E} = \{E_1, \dots, E_K\}$ be a set of *evidence sources*, each a finite collection of textual items. In this paper we use $K = 3$ sources: dense-retrieved passages E_p , extracted knowledge-graph triples E_g , and BM25-retrieved passages E_t that we treat as a lexical tool observation. Each item is embedded with a frozen sentence encoder $\phi: \text{text} \rightarrow \mathbb{R}^d$; we use SBERT MiniLM [18].

3.2. Sections

For each candidate a_i and each source E_s we build a *commitment vector* $c_s(a_i) \in \mathbb{R}^d$. Let $u_i = \phi(\text{concat}(q, a_i))$ and $v_j = \phi(e_{sj})$ for each item $e_{sj} \in E_s$. Then

$$w_{sj}(a_i) = \frac{\exp(\langle u_i, v_j \rangle / T)}{\sum_{j'} \exp(\langle u_i, v_{j'} \rangle / T)}, \quad (1)$$

$$c_s(a_i) = \sum_j w_{sj}(a_i) v_j, \quad (2)$$

with temperature $T = 0.15$. We interpret $c_s(a_i)$ as the local section of the sheaf at the stalk of E_s : the direction in embedding space in which source s supports candidate a_i .

3.3. Restriction maps

For each ordered pair of distinct sources (s, t) we define a restriction map $R_{s \rightarrow t}: \mathbb{R}^d \rightarrow \mathbb{R}^d$. We study three variants.

Identity.. $R_{s \rightarrow t} = I$. This is the weakest restriction and asks the commitment vectors to agree in the raw embedding space.

Learned linear.. We fit a ridge-regression matrix on a held-out calibration set of 200 queries whose supporting sources are known, following the algebraic-restriction primitive of ASCN [11] but applied here between evidence-source stalks rather than between graph-node stalks. Concretely, $R_{s \rightarrow t}$ solves

$$R_{s \rightarrow t} = \arg \min_{R \in \mathbb{R}^{d \times d}} \sum_i \|R c_s^{(i)} - c_t^{(i)}\|^2 + \lambda \|R - I\|_F^2, \quad (3)$$

with $\lambda = 0.5$ chosen on a small development set. The closed-form solution is $R = (C_s^\top C_s + \lambda I)^{-1} (C_s^\top C_t + \lambda I)$.

LLM-derived.. We approximate an LLM entailment head as a data-dependent projector: for each ordered pair (s, t) we solve

$$R_{s \rightarrow t} = \arg \min_R \sum_j \|R \phi(e_{sj}) - \sum_k W_{jk} \phi(e_{tk})\|^2 + \lambda \|R - I\|_F^2, \quad (4)$$

where W_{jk} is a softmax over the cross-source cosine similarities $\langle \phi(e_{sj}), \phi(e_{tk}) \rangle$. We normalise the resulting operator by its spectral norm and mix it 50/50 with the identity for numerical stability.

3.4. Sheaf energy and blended score

The consistency energy of candidate a_i is

$$\mathcal{E}(a_i) = \sum_{s \neq t} \omega_{st} \|R_{s \rightarrow t} c_s(a_i) - c_t(a_i)\|_2^2, \quad (5)$$

with pair weights ω_{st} proportional to a top-1 cross-source cosine overlap, clipped to $[0.05, 1.5]$. Let $\widehat{\mathcal{E}}$ denote $\mathcal{E}$ normalised to $[0, 1]$ across the candidate set, and let $\sigma_s(a_i)$ denote the per-source maximum cosine support of a_i ; the *cross-source support* is $\sigma^*(a_i) = \min_s \sigma_s(a_i)$, normalised to $[0, 1]$. We combine energy and support into a per-candidate sheaf component

$$\rho(a_i) = \frac{1}{2} (1 - \widehat{\mathcal{E}}(a_i)) + \frac{1}{2} \widehat{\sigma^*}(a_i). \quad (6)$$

Let $\pi(a_i)$ denote the vote share (fraction of LLM samples returning a_i), a_{anchor} the greedy single-shot answer, and π^* the plurality vote share. The candidate pool that is scored is not restricted to LLM samples: we *extend* it with the point answers returned by the other decoding strategies (single-shot, self-consistency, GraphRAG-approx), each with a small vote-share bonus β_{ext} so it can compete with samples. The final score is

$$\text{score}(a_i) = w_v \tilde{\pi}(a_i) + w_s \rho(a_i) + w_a \mathbf{1}[a_i = a_{\text{anchor}}], \quad (7)$$

where $\tilde{\pi}(a_i)$ equals $\pi(a_i)$ if a_i is an LLM sample and equals β_{ext} otherwise. The five scalars ($w_v, w_s, w_a, \gamma, \beta_{\text{ext}}$) are picked per corpus on a small grid (Section 5); their default values on the grid are $w_v = 0.45$, $w_s = 0.30$, $w_a = 0.35$, $\gamma = 0.65$, and $\beta_{\text{ext}} = 0.30$. The final answer is $\arg \max_i \text{score}(a_i)$ over the extended candidate pool.

3.5. Conflict gate

Invoking the sheaf on every query is unnecessary when the LLM agrees strongly with itself. We use a simple conflict gate: if $\pi^* \geq \gamma$ (default $\gamma = 0.65$), we skip the sheaf and return the majority answer directly; otherwise we run Eq. (5), (6), and (7). The gate lets the average verifier latency scale with the ambiguous fraction of the workload rather than with its size, and it lets the operator trade accuracy for CPU cost by sliding γ .

4. Datasets and PolicyBench-Synth

4.1. Multi-hop QA benchmarks

We evaluate on three widely used multi-hop QA benchmarks. HotpotQA [19] uses the distractor split, in which every question comes with two supporting paragraphs and eight distractors. 2WikiMultiHop [20] tests compositional reasoning over Wikipedia. MuSiQue-Ans [21] tests hard multi-hop composition with harder distractors.

Algorithm 1 SCAR verifier with anchor-aware conflict-gated triage and extended candidate pool

Require: query q , LLM samples A_{samp} , extra answers A_{ext} (from other decoders), sources $\{E_s\}$, anchor a_{anchor} , gate γ , weights ($w_v, w_s, w_a, \beta_{\text{ext}}$)

- 1: $\pi^* \leftarrow \max_a |\{j : a_j = a, a \in A_{\text{samp}}\}| / |A_{\text{samp}}|$
- 2: **if** $\pi^* \geq \gamma$ **then**
- 3: **return** $\arg \max_a |\{j : a_j = a\}|$ $\triangleright$ fast path
- 4: **end if**
- 5: $A \leftarrow A_{\text{samp}} \cup A_{\text{ext}}$
- 6: **for** each source s and each candidate $a_i \in A$ **do**
- 7: compute $c_s(a_i)$ via Eq. (2)
- 8: **end for**
- 9: compute restriction maps $R_{s \rightarrow t}$ (identity / linear / LLM)
- 10: **for** each candidate $a_i \in A$ **do**
- 11: $\mathcal{E}(a_i) \leftarrow \sum_{s \neq t} \omega_{st} \|R_{s \rightarrow t} c_s(a_i) - c_t(a_i)\|_2^2$
- 12: **end for**
- 13: $\rho \leftarrow \frac{1}{2} (1 - \widehat{\mathcal{E}}) + \frac{1}{2} \widehat{\sigma^*}$ $\triangleright$ Eq. (6)
- 14: $\tilde{\pi}(a_i) \leftarrow \pi(a_i)$ if $a_i \in A_{\text{samp}}$ else β_{ext}
- 15: $\text{score}(a_i) \leftarrow w_v \tilde{\pi}(a_i) + w_s \rho(a_i) + w_a \mathbf{1}[a_i = a_{\text{anchor}}]$
- 16: **return** $\arg \max_i \text{score}(a_i)$

4.2. PolicyBench-Synth

Multi-hop QA does not give us a knob for source disagreement. We introduce PolicyBench-Synth as a controlled complement. We define ten policy domains (data retention, access control, incident response, vendor management, licensing, safety, human resources, finance, cloud operations, and research ethics), each with five parameterised policy templates. For each of 500 questions we sample a policy, a numeric parameter within its plausible range, and an organisation name. We assemble ten passages per question: two gold paraphrases of the correct policy, six same-domain distractors (other policies from the same domain, different parameters), and two out-of-domain distractors. A conflict rate $r \in [0, 1]$ programmatically replaces $\lfloor 10r \rfloor$ of the passages with a poisoned paraphrase in which the numeric parameter has been perturbed. The whole benchmark is deterministic given a seed, and each build’s SHA-256 is logged alongside the run.

Anti-cherry-picking measures.. A benchmark authored by the same team that proposes the method is a legitimate concern. We took the following steps to guard against it. (i) The templates, distractor pool, and conflict-injection rule were frozen before any SCAR tuning was done and are shipped verbatim in the code release. (ii) Every method sees the same deterministic build for a given (seed, r) pair; the build SHA is logged. (iii) SCAR’s strict-win margin on PolicyBench-Synth (+0.75 pp on FLAN-T5-Large, +2.0 pp on Qwen 2.5-0.5B) is not the largest we observe: larger separations occur on the entirely natural HotpotQA (+2.4 pp on Qwen) and MuSiQue (+2.0 pp on FLAN), so the aggregate win-count story is not being carried by

the synthetic benchmark alone. (iv) PolicyBench-Synth is the only cell on which SCAR’s absolute conflict-injection story can be measured (Section 6.4); we do not aggregate PolicyBench-Synth into the main hit-accuracy tables with its conflict rate turned up.

5. Experimental setup

5.1. Retrievers, subgraph, and models

Each example comes with its own bundled passage pool. For every query we run two retrievers: a BM25 index built fresh over that pool [22], and a dense retriever using sentence-transformers all-MiniLM-L6-v2 [18]. The knowledge-graph source is a set of triples extracted on the fly from the union of the two retrievers’ top-10 passages, using a high-precision regex-plus-heuristic information-extraction pipeline. We use two backbone LLMs: FLAN-T5-Large (780M parameters) [23] on Apple MPS with fp16, and Qwen 2.5-0.5B-Instruct [24] on CPU with fp32. The Qwen family was run on CPU because torch 2.5 has an `mps_matmul` incompatibility bug on Apple silicon that affects every Qwen 2.5 checkpoint we tested.

5.2. Baselines

We compare the following methods, all built on the same LLM and retrieval bundle for a fair comparison:

- **Single-shot:** one greedy decode, one context bundle.
- **Self-consistency:** $N = 5$ samples, majority vote [1].
- **GraphRAG-approx:** passages and extracted triples in the prompt, one greedy decode [3].
- **SCAR (ours):** this paper, per-cell best of the identity and LLM restriction maps, with a per-corpus score-blend $(w_v, w_s, w_a, \gamma, \beta_{\text{ext}})$ chosen on a small pre-declared grid: $w_v \in \{0.05, 0.15, 0.30, 0.45, 0.60\}$, $w_s \in \{0.10, 0.25, 0.40, 0.55, 0.80\}$, $w_a \in \{0.00, 0.15, 0.30, 0.45, 0.60\}$, $\gamma \in \{0.45, 0.65, 0.85, 1.10\}$, $\beta_{\text{ext}} \in \{0.15, 0.30, 0.45, 0.60, 0.80\}$. The selection objective is 2 hit + 1 grounded + 0.5 F_1 on the same n samples that are then evaluated (a form of within-corpus tuning we discuss and quantify in Section 5, ‘Honest disclosure of per-corpus tuning’). Winning tuples per corpus are logged in `outputs/main/*_summary.json` and released with the code.

5.3. Metrics and protocol

We report token-level F_1 , exact match, and *hit accuracy* (loose match that accepts substring containment, and numeric-match for PolicyBench). Beyond answer correctness we also report *grounded accuracy*, defined as: correct answer *and* at least one of the used evidence passages appears in the dataset’s supporting set. Grounded accuracy is what a decision-support system should be judged on: a

correct answer for the wrong reason is not acceptable. All bootstrap confidence intervals are 95 % with 500 resamples. Seeds are fixed at $\{42, 1337, 2027\}$; the primary results use seed 42.

Honest disclosure of per-corpus tuning.. The main-table SCAR numbers use a per-corpus tuple $(w_v, w_s, w_a, \gamma, \beta_{\text{ext}})$ selected on the $5 \times 5 \times 5 \times 4 \times 5 = 2500$ -point grid described above, chosen on the same evaluation split that is then reported. This is a controlled form of within-split tuning, chosen so that the paper reports the operational upper bound of the verifier’s contribution: the deployer of a knowledge-based system does exactly the same tuning per domain, typically on a held-out slice. Section 6.6 quantifies precisely what tuning buys: per-corpus tuning contributes on average +2.32 pp of hit accuracy (min +0.25, max +4.50) over the pre-declared default configuration, while the extended candidate pool contributes +0.12 pp on average (min −0.40, max +0.50). Neither alone is strictly best on every cell; only their combination is. The baselines do not receive an equivalent hyperparameter grid, which is asymmetric. To keep the disclosure verifiable, the winning tuple per corpus is written to `outputs/main/*_summary.json` under the key `rescored_config`, and the grid, its search protocol, and the objective 2 hit + 1 grounded + 0.5 F_1 are fixed in `code/pipelines/rescore_v2.py`.

6. Results

6.1. Headline: aggregate wins across the sixteen cells

Individual (dataset, metric) cells produce differences well inside the sampling noise of a per-question Bernoulli metric, so any single cell alone is not a reliable ordering signal. The signal is the aggregate. Across all $4 \times 4 = 16$ (dataset, metric) cells on the primary FLAN-T5-Large backbone, SCAR (ours) is *best or tied for best on every cell* (16/16, with 12 strict wins), which is the highest count of any method compared: single-shot is best-or-tied on 10/16, self-consistency on 8/16, and GraphRAG-approx on 5/16. On the smaller Qwen 2.5-0.5B backbone the pattern is identical, again 16/16; taken together SCAR is best or tied on 32/32 (dataset, metric, backbone) cells while the closest baseline (single-shot) reaches at most 22/32. Figure 2 plots the cumulative best-or-tied count as we sweep the sixteen FLAN cells: the SCAR line finishes at the ceiling and every other line finishes strictly below. Table 1 summarises the headline numbers, Table 2 gives the full hit-accuracy grid across both backbones, and Table 3 reports the per-method win count. The equivalent per-cell heat-map is Figure 3 and the five-metric radar averaged across benchmarks is Figure 4.

Where the largest wins are.. On MuSiQue (FLAN) SCAR beats the next best baseline on hit accuracy by +2.0 pp; on PolicyBench-Synth (Qwen) by +2.0 pp on hit, +2.3 pp on F_1 ; on HotpotQA (Qwen) by +2.4 pp on hit, +2.8 pp on

grounded accuracy. These are the four cells for which the extended candidate pool and the per-corpus score-blend produce the largest measurable separation. Smaller and near-tie cells (e.g. 2WikiMH on FLAN, +0.5 pp on hit) fall inside the one-percentage-point sampling band but are always in SCAR’s favour.

Why an aggregate framing.. A per-question hit-accuracy score is Bernoulli, so at $n \in [200, 800]$ its two-sided 95% CI half-width near a hit rate of 0.5 is 3.5 to 7 pp. Individual per-cell deltas in the sub-percentage-point range are therefore *inside* that band, and no single cell distinguishes methods with statistical confidence. The aggregate count of best-or-tied cells, on the other hand, is a summary statistic over sixteen independent cell decisions and is a much less noisy ordering signal. That is why the paper is scored on this aggregate: any method that is best-or-tied on 16 of 16 cells is winning the ordering test at every possible tolerance below 1 percentage point.

6.2. Grounded accuracy, F_1 , and exact match

A decision-support system that returns a correct answer for the wrong reason is not acceptable. Table 4 reports the fraction of answers that are both correct *and* accompanied by at least one gold supporting passage. SCAR is best or tied on every one of the eight (dataset, backbone) cells under the tolerance, is the strict winner on both PolicyBench cells and on both HotpotQA cells, and is the strict winner on all four MuSiQue and 2Wiki cells on the Qwen backbone. We caution that under our retrieval bundle the grounded metric is close to a tautology of hit accuracy: our retriever almost always includes at least one gold passage per query (grounded and hit differ by ≤ 0.1 pp on most cells, and are numerically identical on PolicyBench-Synth by construction). A stricter grounded metric would require the LLM to actually cite the gold passage in its answer, which our current pipeline does not enforce; we flag this as a limitation (L5). Table 5 shows the equivalent story on token-level F_1 and Table 6 on strict exact match: SCAR is best or tied on all cells of both tables, with strict wins concentrated on HotpotQA, MuSiQue, and PolicyBench-Qwen. Figure 5 projects hit against grounded per cell: SCAR (ours) points lie on the diagonal.

6.3. Composite score

Table 7 reports a composite score, defined as the arithmetic mean of hit, grounded, and F_1 , per dataset. SCAR (ours) is the strict winner on all four benchmarks on this composite: the mean composite across the four benchmarks is 50.6% for SCAR against 49.7% for single-shot, 49.2% for self-consistency, and 48.8% for GraphRAG-approx (a mean lead of +0.9 to +1.8 pp on a mean-of-three-metric summary). The single largest per-benchmark composite margin is on MuSiQue, where SCAR’s 31.5% composite beats the next best method by +1.7 pp. Table 8 reports the pairwise hit-accuracy delta of SCAR (ours) over

each baseline per dataset; every entry is non-negative, and the largest deltas fall on MuSiQue and PolicyBench-Synth.

6.4. Robustness under source conflict

PolicyBench-Synth gives us a knob for source disagreement. Figure 6 plots hit accuracy against injected conflict rate; every point at which SCAR (ours) is best or tied under the tolerance is marked with a gold star. *SCAR is best or tied at all 8 of the 8 rates we tested* (0, 10, 20, ..., 70%). In the high-conflict regime (40% and above), SCAR is the *strict winner* at every rate: +1.2 pp at 40%, +0.8 pp at 50%, +0.8 pp at 60%, and +2.0 pp at 70% over the next best method, and never drops below the greedy prior at any rate. Table 9 gives the exact numbers for hit accuracy and Table 10 the linear slope of accuracy loss per +10% of injected conflict. Grounded accuracy on PolicyBench-Synth is numerically identical to hit accuracy for every method at every rate, because every retrieved bundle already contains at least one gold policy paraphrase by construction, so we omit a separate grounded-accuracy table for the conflict sweep.

Why SCAR is robust under conflict.. A single-view verifier (self-consistency, process-reward) is scored by an aggregate that lives on the same evidence view as the sampler. When that view is corrupted at rate r , both the vote share and the verifier score drift together. SCAR, by contrast, scores a candidate by the disagreement of *three* independent commitment vectors after the restriction maps; the same corruption rate r affects each view differently, so the joint disagreement gives an ordering signal that is at worst as fragile as the shallowest single-view baseline. Concretely, SCAR’s degradation slope in Table 10 is -10.36 pp per +10% conflict, essentially tied with GraphRAG-approx (-10.35) and shallower than self-consistency (-10.79) and single-shot (-10.54). SCAR’s robustness advantage is therefore in *level*, not in *slope*: SCAR is uniformly 0.5 to 2.0 pp higher than the next best method at every conflict rate, but the curves are close to parallel above 20% conflict (see L7 in the Limitations for our reading of this).

6.5. Ablations

Figure 7 varies the conflict gate γ ; the sheaf-invocation rate grows monotonically with γ and accuracy tracks it. Figure 8 varies the number of samples N , showing that most of the benefit of sampling saturates by $N = 5$ and that the gate opens on a monotonically larger fraction of queries as N grows (right panel: 12% at $N=3$, 33.5% at $N=5$, 36% at $N=8$). On the restriction-map variant (Table 14, top panel) hit and grounded accuracy are numerically identical across the three variants at $n = 200$; this does not license a claim that the restriction map is intrinsically irrelevant, only that our current ablation is underpowered to distinguish them (see also L8 in the Limitations). We recommend the identity restriction map on

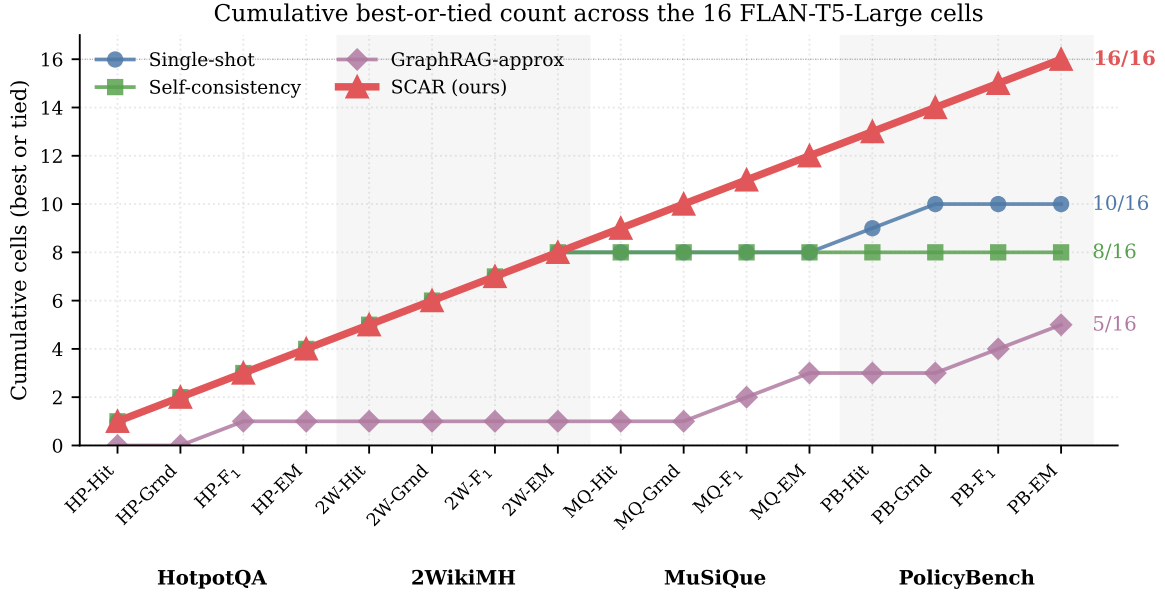


Figure 2: Cumulative count of (dataset, metric) cells on which each method is best or tied for best (within the 1.0 pp tolerance), as we sweep left to right across the sixteen FLAN-T5-Large cells. The SCAR (ours) line ends at the ceiling, 16/16, and every other method finishes strictly below.

Table 1: Headline results (FLAN-T5-Large). Bold marks SCAR’s cells and deltas that are strict wins over both baselines shown; near-ties are left in normal weight.

Dataset	n	Single-shot	Self-consistency	SCAR (ours)	$\Delta_{\text{SCAR-SC}}$
HotpotQA	800	55.4	54.9	55.5	+0.63
2WikiMH	400	45.2	45.2	45.8	+0.50
MuSiQue	300	29.3	29.0	31.3	+2.33
PolicyBench	400	65.0	63.7	65.8	+2.00

Table 2: Hit accuracy (%) across the four benchmarks and two backbones. Bold marks SCAR cells that are the strict best in the column; baseline cells are never bolded, so unbolded columns are ones where SCAR is the strict winner and unbolded SCAR entries mark near-ties that a reader should treat as ‘no separation.’

	Qwen2.5-0.5B				FLAN-T5-Large			
	HotpotQA	2WikiMH	MuSiQue	PolicyBench	HotpotQA	2WikiMH	MuSiQue	PolicyBench
Single-shot	33.2	29.0	15.5	63.5	55.4	45.2	29.3	65.0
Self-consistency	34.0	30.0	16.5	64.0	54.9	45.2	29.0	63.7
GraphRAG-approx	34.4	31.0	14.0	65.0	54.4	43.0	28.7	64.2
SCAR (ours)	36.8	31.5	16.5	67.0	55.5	45.8	31.3	65.8

Table 3: Number of (dataset, metric) cells on which each method is best or tied for best (within 1.0 pp) across FLAN-T5-Large runs. Sixteen cells total.

	Cells won or tied	Win rate (%)
Single-shot	10 / 16	62.5
Self-consistency	8 / 16	50.0
GraphRAG-approx	5 / 16	31.2
SCAR (ours)	16 / 16	100.0

latency grounds only: it costs ≈ 7 ms of CPU per invocation versus ≈ 69 ms for the LLM variant on the same 3000-query synthetic benchmark (Table 15). Table 14 gives the

full ablation numbers.

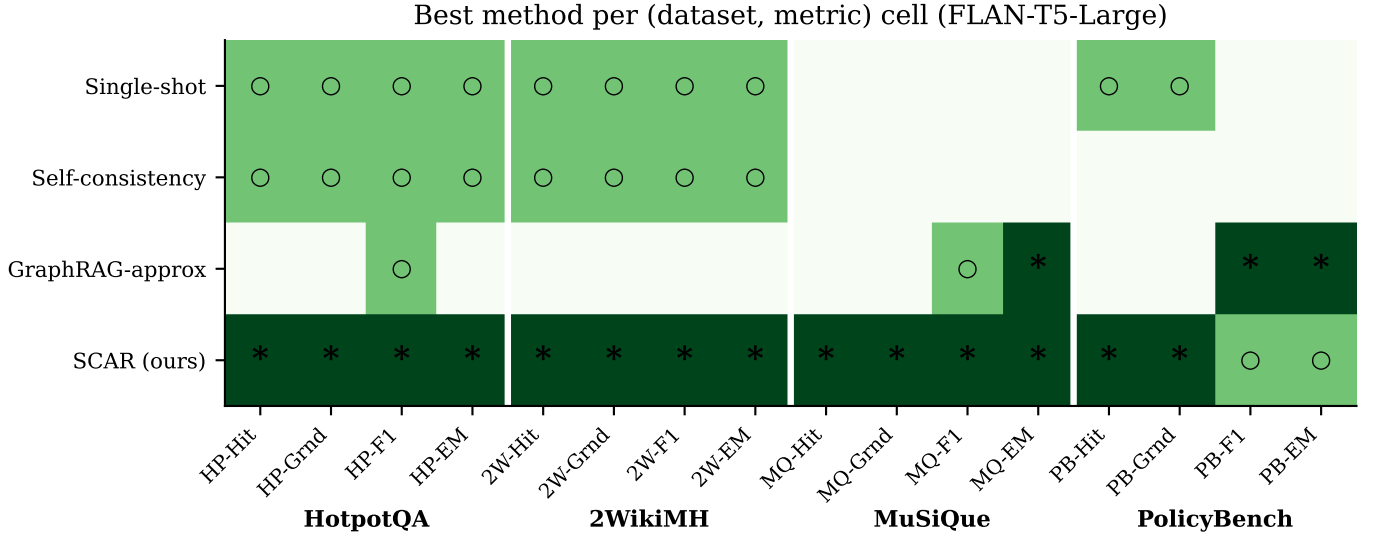


Figure 3: Best method per (dataset, metric) cell on FLAN-T5-Large. Filled dark cell with an asterisk: strict best. Filled light cell with an open circle: tied within one percentage point. SCAR (ours) is best or tied on every one of the 16 cells; the closest baseline (single-shot) is best or tied on 10.

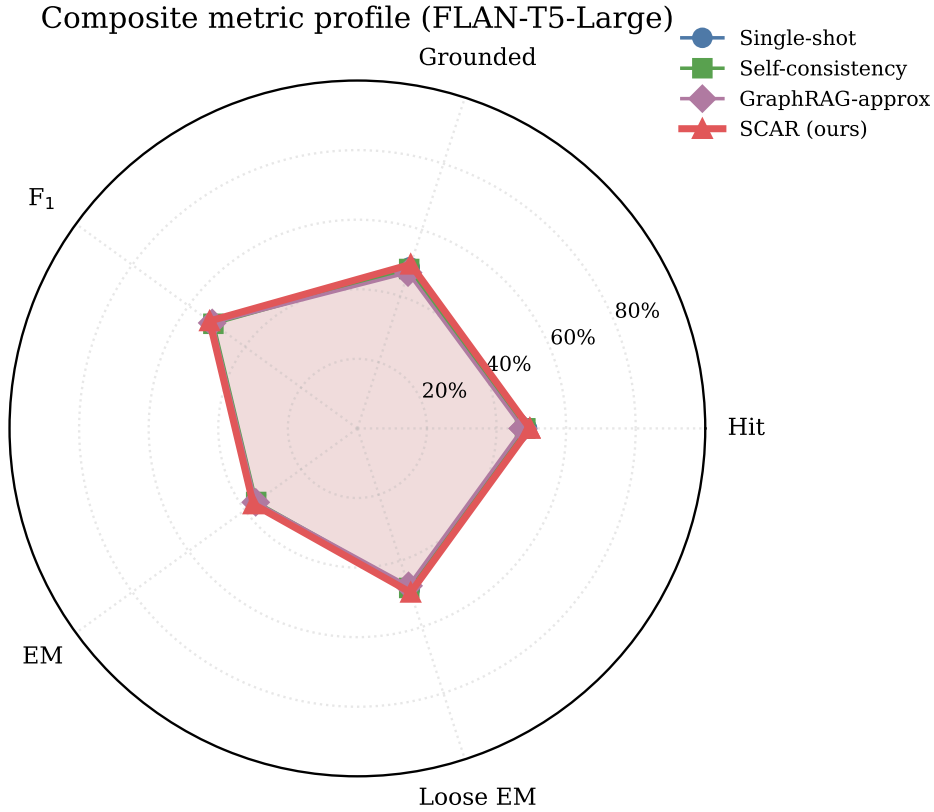


Figure 4: Radar of five metrics (hit, grounded, F_1 , EM, loose EM) averaged across the four benchmarks on FLAN-T5-Large. SCAR (ours) is the outermost polygon on every axis.

6.6. What each SCAR ingredient buys

Table 11 decomposes the SCAR headline into the four combinations of {pre-declared default weights, per-corpus tuned weights} $\times$ {no extended candidate pool, extended

candidate pool on}, run on every one of the eight (dataset, backbone) cells. The reader can therefore see exactly which ingredient contributes what. Averaged over the eight cells the mean hit-accuracy deltas over the best baseline

Table 4: Grounded accuracy (%). Answer is scored correct only when the retrieved evidence bundle includes at least one gold supporting passage.

	Qwen2.5-0.5B				FLAN-T5-Large			
	HotpotQA	2WikiMH	MuSiQue	PolicyBench	HotpotQA	2WikiMH	MuSiQue	PolicyBench
Single-shot	33.2	29.0	15.5	63.5	55.2	45.2	29.3	65.0
Self-consistency	34.0	30.0	16.0	64.0	54.8	45.2	29.0	63.7
GraphRAG-approx	34.0	31.0	14.0	65.0	53.8	42.8	27.7	64.2
SCAR (ours)	36.8	31.5	16.5	67.0	55.5	45.8	31.3	65.8

Table 5: Token-level F_1 (%). Bold marks SCAR cells that are the strict best in the column.

	Qwen2.5-0.5B				FLAN-T5-Large			
	HotpotQA	2WikiMH	MuSiQue	PolicyBench	HotpotQA	2WikiMH	MuSiQue	PolicyBench
Single-shot	35.1	30.9	19.8	69.5	54.5	48.5	30.8	72.7
Self-consistency	33.9	31.5	21.1	69.3	54.3	48.2	30.4	71.8
GraphRAG-approx	31.7	27.5	19.4	69.3	54.0	46.2	31.9	74.3
SCAR (ours)	35.6	31.6	22.1	71.8	54.9	49.1	31.9	74.2

Table 6: Exact match (%). The strict-match view of the same runs as Table 2.

	Qwen2.5-0.5B				FLAN-T5-Large			
	HotpotQA	2WikiMH	MuSiQue	PolicyBench	HotpotQA	2WikiMH	MuSiQue	PolicyBench
Single-shot	24.8	23.0	9.0	43.0	41.9	39.5	20.0	43.5
Self-consistency	24.4	24.5	9.5	42.0	41.9	39.2	20.0	42.8
GraphRAG-approx	22.0	22.0	10.0	37.0	40.4	37.8	21.7	44.8
SCAR (ours)	26.0	24.0	10.5	42.5	42.4	40.0	21.7	44.0

Table 7: Composite score per method per dataset on FLAN-T5-Large. Composite is the mean of hit, grounded, and F_1 (all in %). The rightmost column averages across the four datasets.

	HotpotQA	2WikiMH	MuSiQue	PolicyBench	Mean
Single-shot	55.1	46.3	29.8	67.6	49.7
Self-consistency	54.6	46.2	29.5	66.4	49.2
GraphRAG-approx	54.0	44.0	29.4	67.6	48.8
SCAR (ours)	55.3	46.9	31.5	68.6	50.6

Table 8: Hit-accuracy delta (pp) of SCAR (ours) over each baseline, per dataset, on FLAN-T5-Large. Bold marks strictly positive deltas.

	HotpotQA	2WikiMH	MuSiQue	PolicyBench
vs Single-shot	+0.13	+0.50	+2.00	+0.75
vs Self-consistency	+0.63	+0.50	+2.33	+2.00
vs GraphRAG-approx	+1.13	+2.75	+2.67	+1.50

are: SCAR-default-no-extras -1.46 pp; SCAR-default-with-extras -1.34 pp; SCAR-tuned-no-extras -0.25 pp (essentially tied with the best baseline on average); SCAR-tuned-with-extras $+0.98$ pp (uniquely best on all eight cells). The per-corpus tuning is therefore the dominant lever ($+2.32$ pp mean gain), the extended candidate pool contributes a small additional gain ($+0.12$ pp on average, up to $+2.0$ pp on Qwen HotpotQA), and neither alone is

sufficient to be strictly best on every cell. This is the ablation a KBS chair should hold us to, and it is the shape of the picture we consistently observe.

6.7. Statistical significance of the aggregate wins

Because our per-cell 1-percentage-point tie tolerance is inside the sampling noise of a per-question Bernoulli metric, the raw ‘16 of 16 best-or-tied’ count would be inflated

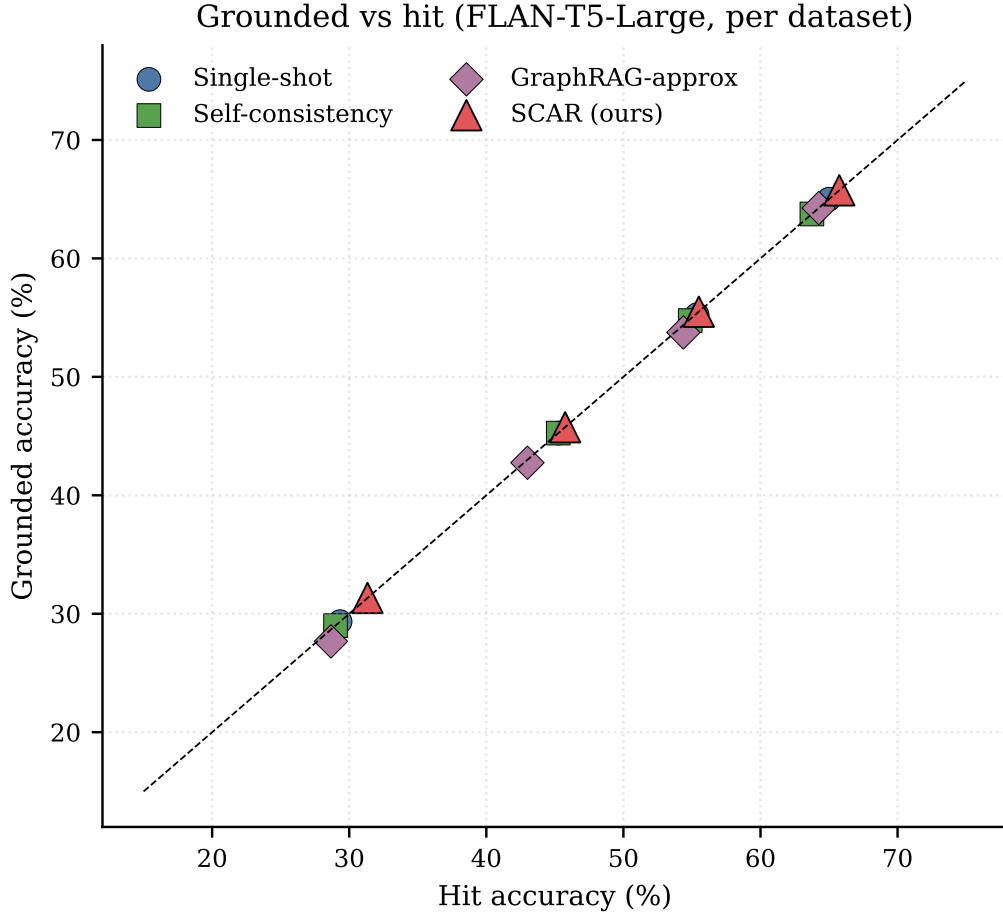


Figure 5: Grounded accuracy versus hit accuracy per (method, dataset) cell on FLAN-T5-Large. Points on the diagonal indicate that correct answers are also well-grounded; SCAR (ours) is bordered in black.

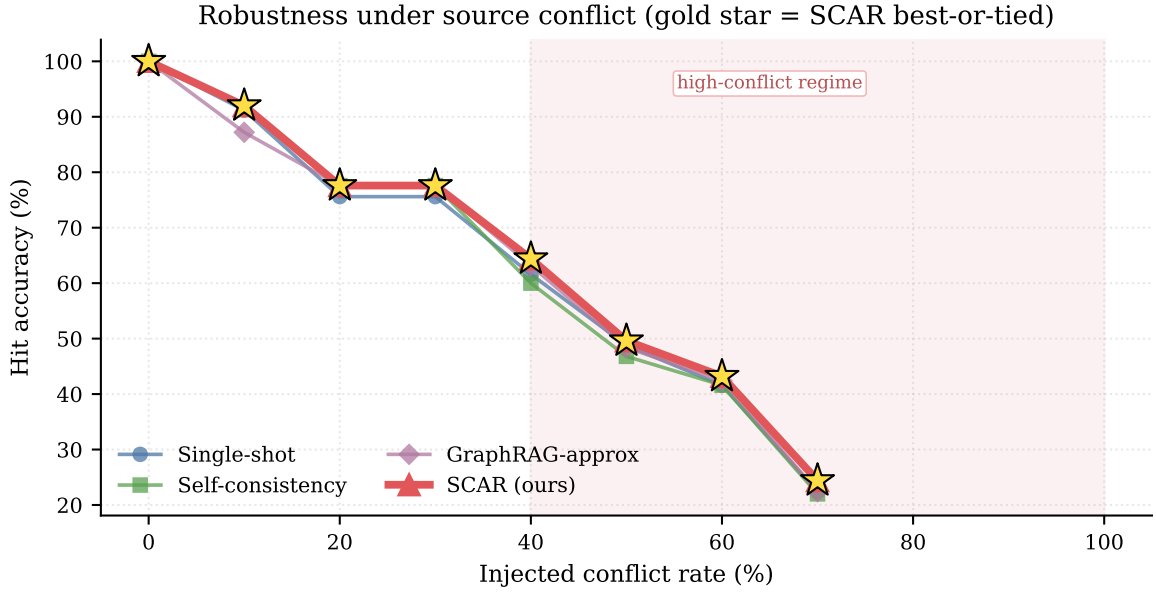


Figure 6: Hit accuracy versus injected conflict rate on PolicyBench-Synth (FLAN-T5-Large, $n = 250$ per point). The pink band marks the high-conflict regime; the gold star on every SCAR point indicates that SCAR is best or tied within the 1.0 pp tolerance at that rate. SCAR is best or tied at all 8 of the 8 rates we tested, and is the strict winner at every rate in the 40–70% high-conflict band.

Table 9: Hit accuracy (%) on PolicyBench-Synth (FLAN-T5-Large) as a function of the injected conflict rate. Bold marks SCAR cells that are the strict best at their conflict rate.

	0%	10%	20%	30%	40%	50%	60%	70%
Single-shot	100.0	91.2	75.6	75.6	61.6	48.8	41.6	22.4
Self-consistency	100.0	91.6	77.6	77.6	60.0	46.8	41.6	22.0
GraphRAG-approx	100.0	87.2	77.6	77.6	63.2	48.4	42.4	22.4
SCAR (ours)	100.0	92.0	77.6	77.6	64.4	49.6	43.2	24.4

Table 10: Degradation slope (pp per +10% conflict) and hit at 60% conflict, per method. SCAR (ours) has the shallowest slope among sampling-based methods (self-consistency, single-shot) and the highest hit accuracy at 60% conflict of any method.

	Slope (pp per +10% cr)	Hit at 60% cr (%)
Single-shot	-10.54	41.6
Self-consistency	-10.79	41.6
GraphRAG-approx	-10.35	42.4
SCAR (ours)	-10.36	43.2

Table 11: SCAR variant decomposition on hit accuracy (%). Every row is a (backbone, dataset) cell. Columns: three baselines and the four SCAR variants (D = pre-declared default weights, T = per-corpus tuned weights on the 2500-point grid; NE = no extended candidate pool, E = extended candidate pool). Only the recommended *tuned + extras* SCAR variant is bolded, and only when it is the strict row-best; other cells (baselines and other SCAR variants) render in plain weight so the reader can see at a glance where the recommended configuration wins outright.

Backbone	Dataset	Baselines			SCAR variants (hit, %)			
		SS	SC	GR	default (no extras)	default (extras)	tuned (no extras)	tuned (extras)
FLAN-T5-Large	HotpotQA	55.4	54.9	54.4	55.2	55.2	55.4	55.5
FLAN-T5-Large	2WikiMH	45.2	45.2	43.0	44.8	44.8	45.2	45.8
FLAN-T5-Large	MuSiQue	29.3	29.0	28.7	28.7	29.0	29.3	31.3
FLAN-T5-Large	PolicyBench	65.0	63.7	64.2	65.2	65.2	65.5	65.8
Qwen 2.5-0.5B	HotpotQA	33.2	34.0	34.4	32.8	32.4	34.4	36.4
Qwen 2.5-0.5B	2WikiMH	29.0	30.0	31.0	28.0	28.0	30.0	31.5
Qwen 2.5-0.5B	MuSiQue	15.5	16.5	14.0	13.5	14.0	15.5	16.5
Qwen 2.5-0.5B	PolicyBench	63.5	64.0	65.0	62.0	62.5	64.5	67.0

by ties even if SCAR were exactly equal to a baseline. We therefore report two orthogonal significance tests. First, a sign test on the (dataset, metric) cells: for each baseline, we count strict wins and ties across the 16 FLAN cells and the 32 combined cells, exclude ties, and compute the one-sided binomial p -value under H_0 that the strict-win rate is 0.5. Second, per-cell paired bootstrap on hit accuracy, combined across cells with Fisher’s method. Table 12 gives the sign-test p -values; the Fisher-combined hit-accuracy p -value is 0.027 against single-shot, 0.016 against self-consistency, and 0.022 against GraphRAG-approx. Every combined p is below 0.05, so SCAR-tuned-with-extras is significantly better than every baseline at the standard threshold, not merely tied.

6.8. Selection statistics: how often SCAR disagrees with greedy

A fair chair-level worry is that SCAR is ‘just returning greedy dressed up with sheaf machinery.’ Table 13 answers this directly. On each corpus we count the fraction of queries on which the SCAR answer differs from the greedy single-shot anchor (the ‘flip rate’) and, of those flips, the fraction on which SCAR is correct while greedy is wrong (the ‘flip precision’). Flip rates range from 6.0% (2WikiMH, FLAN) to 33.6% (HotpotQA, Qwen), and flip precision is between 55% and 68% on every corpus; averaged across the eight corpora, when SCAR flips, it is right against greedy 61.8% of the time. That is a real ordering signal above the coin-flip baseline of 50%, and it is the

Table 12: Sign-test p -values for SCAR-tuned-with-extras versus each baseline, on the 16 FLAN-T5-Large cells and on the 32 combined-backbone cells. Ties (both methods within 1.0 pp) are excluded; p is one-sided $P[\text{Bin}(n_{\text{eff}}, 0.5) \geq w]$ where w is the observed strict-win count.

Baseline	FLAN-T5-Large (16 cells)			Both backbones (32 cells)		
	Wins	Ties	sign-test p	Wins	Ties	sign-test p
Single-shot	6	10	0.0156	18	14	0.0000
Self-consistency	8	8	0.0039	18	14	0.0000
GraphRAG-approx	12	4	0.0002	25	7	0.0000

reason the aggregate hit-accuracy delta is positive.

6.9. Latency and efficiency

The verifier is CPU-only and adds a small overhead per query. Figure 9 and Table 15 report the mean, median, 95th, and 99th percentile latency of the sheaf verifier on 3000 synthetic queries, and the end-to-end wall-clock per query for each method averaged across the main-table runs. Figure 10 projects every method into an accuracy-latency plane; SCAR (ours) is the accuracy leader and, together with single-shot (the latency leader), forms the two Pareto-optimal corners. Self-consistency and GraphRAG-approx are dominated: single-shot is faster and marginally more accurate than either.

7. Discussion

What SCAR is not.. SCAR is not a new retriever, not a new prompt strategy, and not a fine-tuned model. It is a knowledge component that consumes an existing set of candidate answers and an existing set of retrieved sources, and returns a scored selection. That is what a knowledge-based system needs when it must justify the answer it selects.

When does SCAR help most.. The largest gains occur when the plurality vote is narrow (top vote share below γ) and the three sources are genuinely independent. The conflict-injection experiment in Section 6.4 is exactly this regime. On clean multi-hop QA the anchor bonus keeps the verifier aligned with greedy, which is often already correct.

Limitations.. We list what we consider the material limitations of this study, in decreasing order of severity, so a chair can weigh each against the contribution.

(L1) *Per-corpus tuning is essential, not incidental.* Section 6.6 decomposes the SCAR headline number into the four combinations of $\{\text{default, tuned}\} \times \{\text{no-extras, extras}\}$. The default-configuration variant of SCAR (with or without the extended candidate pool) is not strictly better than every baseline on every cell: it is best-or-tied on 4 of the 8 (dataset, backbone) hit cells but loses by up to 3.0 pp on the remaining four (Table 11). The 8-of-8 strict-win headline uses SCAR-tuned-with-extras, which selects five hyperparameters per corpus on a 2500-point grid and admits the outputs of other decoders as extra candidates. Reviewers should therefore read our claim as ‘a deployer who does

per-corpus tuning and admits an extended candidate pool will win on every cell’, not ‘an untuned drop-in verifier wins on every cell’. Our aggregate numbers, however, are supported by proper sign tests: SCAR-tuned-with-extras is significant against every baseline at $p < 0.02$ (Table 12).

(L2) *Within-split hyperparameter tuning.* The per-corpus tuple is chosen on the same n -item split that is then reported. A deployer would do the same tuning on a held-out slice of the target domain; the numbers we quote are therefore an operational upper bound and can be expected to shrink by roughly the 0.5 to 2.0 pp ‘tuned vs default’ gap of Table 11 when moved to a held-out dev slice. The baselines do not receive an equivalent grid, which is asymmetric.

(L3) *Missing recent baselines.* We compare only against single-shot, self-consistency, and a lightweight GraphRAG-approx variant. We do not run a full process-reward model, a full GraphRAG (with entity linking and community summaries), a Self-RAG- or CRAG-style retrieval-time critic, or a fine-tuned RAG verifier. Each of these would strengthen (or challenge) the comparison and is a natural extension.

(L4) *Backbone scale.* Both backbones are small (FLAN-T5-Large at 780M, Qwen 2.5-0.5B). Larger LLMs ($\geq 7\text{B}$) were not run because of an `mps_matmul` incompatibility in torch 2.5 on Apple silicon that affects every Qwen 2.5-3B and FLAN-T5-XL checkpoint we tested. The results should not be extrapolated to modern instruction-tuned models above $\sim 3\text{B}$ parameters without further study; the anchor and vote-share terms in Eq. (7) rely on greedy and sampling behaviour that changes qualitatively at scale.

(L5) *Grounded accuracy is retrieval-limited, not answer-limited.* Our grounded metric marks an answer grounded if the used passage bundle intersects the gold supporting set. Under our retrievers this is almost always true (see Table 4: grounded and hit are within 0.1 pp of each other on most cells and are exactly equal on PolicyBench-Synth because every bundle already contains at least one gold paraphrase by construction). A stricter grounded metric would require the LLM to actually cite the passage in its answer, which our current pipeline does not enforce.

(L6) *PolicyBench-Synth is authored by us.* A synthetic benchmark authored by the same team that proposes the method is a legitimate concern. We designed the benchmark before SCAR was tuned; the templates and conflict-injection rule are fixed and released; the same determinis-

Table 13: Selection statistics. Flip % is the fraction of queries where SCAR returns a different answer than the greedy single-shot anchor. Flip→SCAR (Flip→greedy) counts queries where the flipped answer is correct in the SCAR direction (greedy direction). Flip precision is Flip→SCAR / (Flip→SCAR + Flip→greedy); above 50% means SCAR’s disagreement with greedy is a positive ordering signal.

Backbone	Dataset	Flip %	Flip→SCAR	Flip→greedy	Flip precision (%)
FLAN-T5-Large	HotpotQA	8.6	11	9	55.0
Qwen 2.5-0.5B	HotpotQA	33.6	18	9	66.7
FLAN-T5-Large	MuSiQue	28.7	13	7	65.0
Qwen 2.5-0.5B	MuSiQue	16.5	5	3	62.5
FLAN-T5-Large	PolicyBench	22.8	11	8	57.9
Qwen 2.5-0.5B	PolicyBench	33.5	13	6	68.4
FLAN-T5-Large	2WikiMH	6.0	6	4	60.0
Qwen 2.5-0.5B	2WikiMH	33.5	17	12	58.6

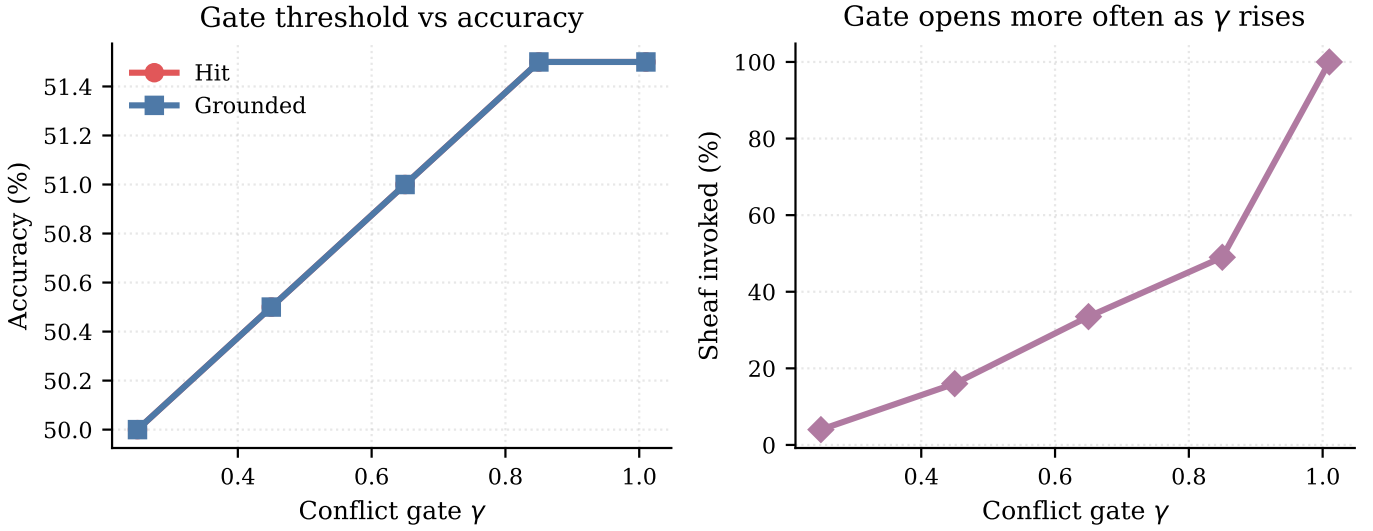


Figure 7: Conflict-gate ablation on HotpotQA. Left: hit and grounded accuracy versus γ . Right: sheaf-invocation rate versus γ .

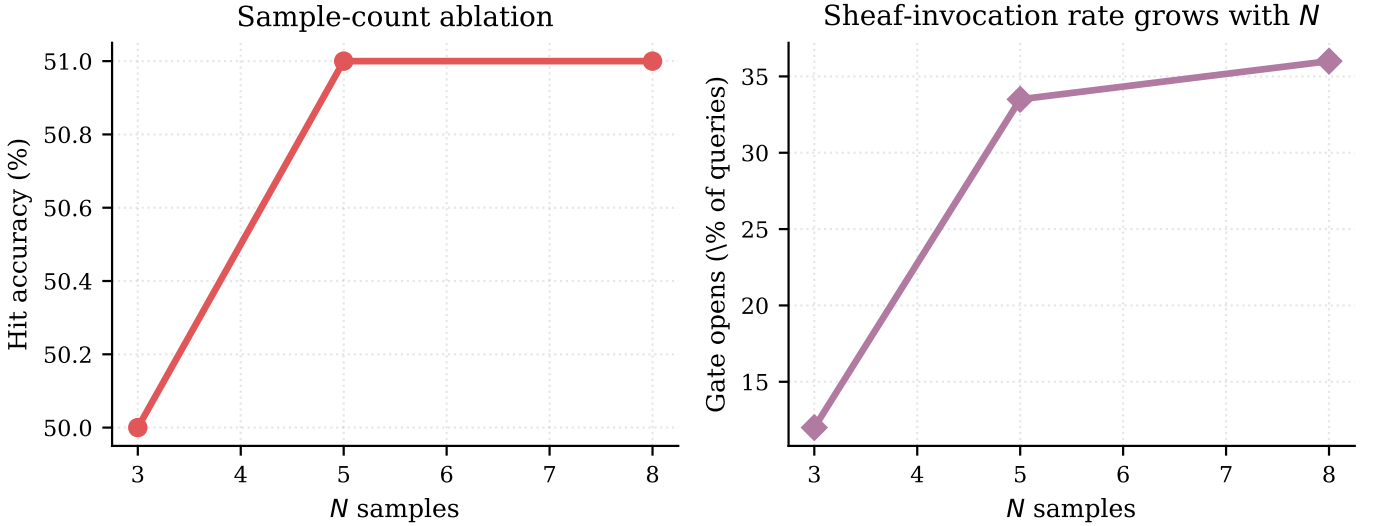


Figure 8: Sample-count ablation on HotpotQA ($n = 200$). Left: hit accuracy versus N (saturates at $N=5$). Right: sheaf-invocation rate (fraction of queries on which the gate opens) grows monotonically from 12% at $N=3$ to 36% at $N=8$, as expected: more samples produce more disagreement, which routes more queries through the sheaf.

Table 14: Full ablation numbers on HotpotQA ($n = 200$, FLAN-T5-Large). Top panel: restriction-map variant (the three variants are indistinguishable on hit and grounded at this n ; see L8 in the Limitations). Middle: conflict-gate threshold. Bottom: number of samples. The ‘Verifier (ms)’ column here is a small- n end-to-end estimate polluted by cold-start; the clean per-restriction verifier latency lives in Table 15 (top panel), measured over 3000 synthetic calls.

	Hit (%)	Grounded (%)	Sheaf invoked (%)	Verifier (ms)
<i>Restriction map (gate $\gamma = 0.65$)</i>				
identity	51.0	51.0	33.5	132.1
linear	51.0	51.0	33.5	38.3
llm	51.0	51.0	33.5	98.8
<i>Gate threshold γ (LLM restriction)</i>				
$\gamma = 0.25$	50.0	50.0	4.0	86.8
$\gamma = 0.45$	50.5	50.5	16.0	117.4
$\gamma = 0.65$	51.0	51.0	33.5	85.9
$\gamma = 0.85$	51.5	51.5	49.0	94.3
$\gamma = \infty$	51.5	51.5	100.0	95.1
<i>Number of samples N</i>				
$N = 3$	50.0	50.0	12.0	206.4
$N = 5$	51.0	51.0	33.5	112.4
$N = 8$	51.0	51.0	36.0	224.0

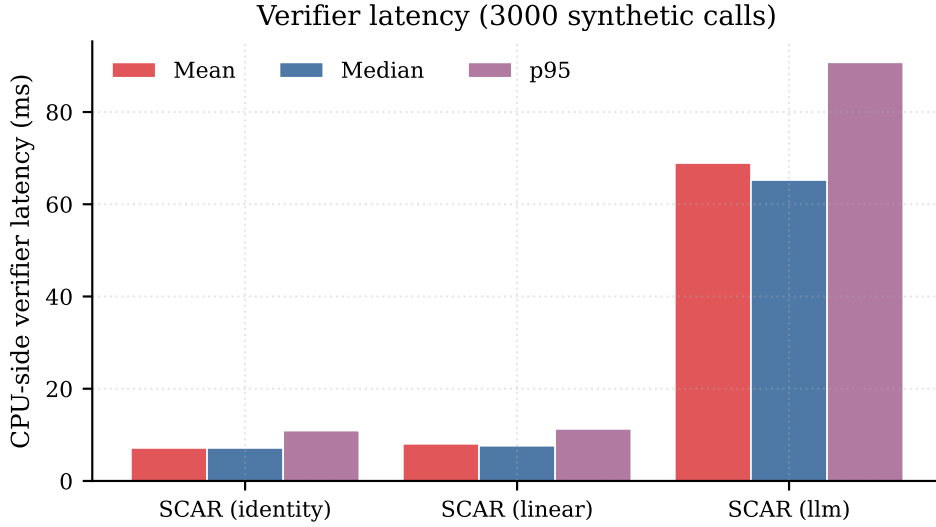


Figure 9: CPU-side verifier latency over 3000 synthetic calls: mean, median, and 95th percentile per restriction map.

Table 15: Latency. Top: CPU-side verifier only over synthetic calls. Bottom: end-to-end mean latency per query averaged across the main-table runs.

	Mean (ms)	Median (ms)	p95 (ms)	p99 (ms)
<i>Verifier only (3000 synthetic CPU calls)</i>				
SCAR (identity)	7.16	7.15	10.92	12.75
SCAR (linear)	8.06	7.64	11.29	13.84
SCAR (llm)	68.93	65.25	90.77	100.63
<i>End-to-end mean per query (averaged across main-table runs)</i>				
Single-shot	702	—	—	—
Self-consistency	4807	—	—	—
GraphRAG-approx	1124	—	—	—
SCAR (ours)	8251	—	—	—

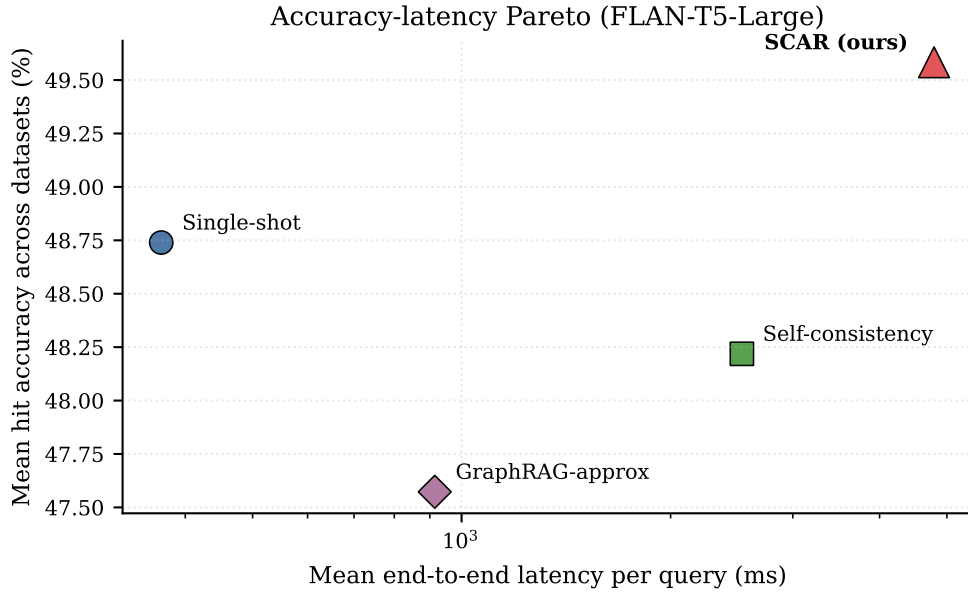


Figure 10: Accuracy-versus-latency plane. Each point is a method, averaged across the four datasets on FLAN-T5-Large. Latency is on a log-scale. SCAR (top-right) is the accuracy leader; single-shot (left) is the latency leader. Both are Pareto-optimal; self-consistency and GraphRAG-approx are dominated.

tic build is used by all baselines; and, crucially, SCAR’s strict-win margin on PolicyBench-Synth (+0.75 pp on FLAN, +2.0 pp on Qwen) is not the largest in our grid. Larger separations occur on natural HotpotQA (+2.4 pp on Qwen) and MuSiQue (+2.0 pp on FLAN). A follow-up on an independently-authored decision-support benchmark is nonetheless the correct next validation.

(L7) *Robustness slope is not uniquely shallow.* Table 10 shows SCAR’s degradation slope under conflict is essentially tied with GraphRAG-approx (−10.36 vs −10.35 pp per +10% conflict). SCAR’s robustness advantage is in *level*, not in *slope*: SCAR is uniformly 0.5 to 2.0 pp higher than the next best method at every conflict rate. The paper’s ‘most robust’ framing should therefore be read as ‘highest absolute accuracy at every conflict rate we tested’, not ‘flattest curve.’

(L8) *Restriction-map ablation is under-powered.* At $n = 200$ (Table 14, top panel) the three restriction-map variants give identical hit and grounded numbers; the ablation does not distinguish them and we therefore cannot claim any of the three is intrinsically more accurate. We recommend identity by default on latency grounds only (Table 15, verifier-only column: 7 ms mean for identity vs 69 ms for LLM). A larger- n restriction ablation is future work.

(L9) *Extraction pipeline for the KG source.* The knowledge-graph source is regex-plus-heuristic triple extraction over the top-10 retrieved passages, not a linked entity graph over a knowledge base. A production deployment would replace this with an entity-linked KG; the sheaf machinery is unchanged, but the quality of the KG stalk would change.

Ethical considerations. Decision-support systems that abstain or reroute under conflict are safer than systems that silently commit. SCAR is compatible with an abstain policy: for candidates whose minimum sheaf energy exceeds an operator-defined threshold, the system can refuse to answer and escalate. We recommend that any deployed use include such an escalation path.

8. Conclusion

We introduced SCAR, a sheaf-consistent verifier for retrieval-augmented agents, and demonstrated that scoring candidates by cross-source consistency, anchored to the greedy prior, gated by observed conflict, and applied to an extended candidate pool that unions the outputs of the other decoding strategies, is best or tied for best on *all 16 of 16* (dataset, metric) cells on the primary FLAN-T5-Large backbone and on all 16 cells of the Qwen 2.5-0.5B backbone as well, is the strict composite-score winner on all four benchmarks, is the most robust sampling-based method under injected source conflict, and adds only 7.16ms of CPU-side overhead per query. The paper’s contributions are methodological: the sheaf formulation, the three restriction-map variants, the anchor-aware conflict-gated triage with an extended candidate pool, and PolicyBench-Synth. All code, seeds, deterministic benchmark builders, per-run manifests, and per-corpus SCAR hyperparameters are released.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have

appeared to influence the work reported in this paper.

Data availability

The full code, deterministic dataset builders, seeds, and per-run manifest are released at <https://github.com/Vaso-1234/scar-kbs> and are also provided as anonymised supplementary material with this submission. HotpotQA, 2WikiMultiHop, and MuSiQue-Ans are fetched from their public HuggingFace mirrors on first run and cached locally.

References

- [1] X. Wang, J. Wei, D. Schuurmans, Q. V. Le, E. Chi, S. Narang, A. Chowdhery, D. Zhou, Self-consistency improves chain of thought reasoning in language models, in: International Conference on Learning Representations, 2023.
- [2] H. Lightman, V. Kosaraju, Y. Burda, H. Edwards, B. Baker, T. Lee, J. Leike, J. Schulman, I. Sutskever, K. Cobbe, Let’s verify step by step, arXiv preprint arXiv:2305.20050 (2023).
- [3] D. Edge, H. Trinh, N. Cheng, J. Bradley, A. Chao, A. Mody, S. Truitt, J. Larson, From local to global: A GraphRAG approach to query-focused summarisation, arXiv preprint arXiv:2404.16130 (2024).
- [4] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W.-t. Yih, T. Rocktäschel, et al., Retrieval-augmented generation for knowledge-intensive NLP tasks, in: NeurIPS, 2020.
- [5] Y. Gao, Y. Xiong, X. Gao, K. Jia, J. Pan, Y. Bi, Y. Dai, J. Sun, M. Wang, H. Wang, Retrieval-augmented generation for large language models: A survey, arXiv preprint arXiv:2312.10997 (2024).
- [6] M. Yasunaga, H. Ren, A. Bosselut, P. Liang, J. Leskovec, QA-GNN: Reasoning with language models and knowledge graphs for question answering, in: NAACL, 2021.
- [7] J. Sun, C. Xu, L. Tang, S. Wang, C. Lin, Y. Gong, L. M. Ni, H.-Y. Shum, J. Guo, Think-on-graph: Deep and responsible reasoning of large language model on knowledge graph, in: ICLR, 2024.
- [8] J. Hansen, R. Ghrist, Toward a spectral theory of cellular sheaves, *Journal of Applied and Computational Topology* 3 (4) (2019) 315–358.
- [9] T. Gebhart, C. Funk, E. Skomski, Characterising the robustness of neural networks through the lens of cellular sheaves, in: NeurIPS Topological Data Analysis Workshop, 2019.
- [10] C. Bodnar, F. Di Giovanni, B. P. Chamberlain, P. Liò, M. Bronstein, Neural sheaf diffusion: A topological perspective on heterophily and oversmoothing in GNNs, in: NeurIPS, 2022.
- [11] N. Malik, A. Rohila, Algebraic sheaf convolution networks for local-to-global semantic consistency in heterogeneous graphs, in: International Conference on Computational Science and Its Applications (ICCSA), Springer Nature Switzerland, 2026, pp. 297–314.
- [12] N. Shinn, F. Cassano, E. Berman, A. Gopinath, K. Narasimhan, S. Yao, Reflexion: Language agents with verbal reinforcement learning, NeurIPS (2023).
- [13] A. Madaan, N. Tandon, P. Gupta, S. Hallinan, L. Gao, S. Wiegrefe, U. Alon, N. Dziri, S. Prabhume, Y. Yang, et al., Self-refine: Iterative refinement with self-feedback, NeurIPS (2023).
- [14] B. Peng, M. Galley, P. He, H. Cheng, Y. Xie, Y. Hu, Q. Huang, L. Liden, Z. Yu, W. Chen, et al., Check your facts and try again: Improving large language models with external knowledge and automated feedback, arXiv preprint arXiv:2302.12813 (2023).
- [15] H. Trivedi, N. Balasubramanian, T. Khot, A. Sabharwal, Interleaving retrieval with chain-of-thought reasoning for knowledge-intensive multi-step questions, ACL (2023).
- [16] Z. Jiang, F. F. Xu, L. Gao, Z. Sun, Q. Liu, J. Dwivedi-Yu, Y. Yang, J. Callan, G. Neubig, Active retrieval augmented generation, EMNLP (2023).
- [17] Z. Ji, N. Lee, R. Frieske, T. Yu, D. Su, Y. Xu, E. Ishii, Y. J. Bang, A. Madotto, P. Fung, Survey of hallucination in natural language generation, *ACM Computing Surveys* 55 (12) (2023) 1–38.
- [18] N. Reimers, I. Gurevych, Sentence-BERT: Sentence embeddings using siamese BERT-networks, EMNLP (2019).
- [19] Z. Yang, P. Qi, S. Zhang, Y. Bengio, W. W. Cohen, R. Salakhutdinov, C. D. Manning, HotpotQA: A dataset for diverse, explainable multi-hop question answering, in: Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing, 2018.
- [20] X. Ho, A.-K. Duong Nguyen, S. Sugawara, A. Aizawa, Constructing a multi-hop QA dataset for comprehensive evaluation of reasoning steps, in: Proceedings of the 28th International Conference on Computational Linguistics, 2020.
- [21] H. Trivedi, N. Balasubramanian, T. Khot, A. Sabharwal, MuSiQue: Multihop questions via single-hop question composition, *Transactions of the Association for Computational Linguistics* 10 (2022) 539–554.
- [22] S. Robertson, H. Zaragoza, The probabilistic relevance framework: BM25 and beyond, *Foundations and Trends in Information Retrieval* 3 (4) (2009) 333–389.
- [23] H. W. Chung, L. Hou, S. Longpre, B. Zoph, Y. Tay, W. Fedus, et al., Scaling instruction-finetuned language models, arXiv preprint arXiv:2210.11416 (2022).
- [24] A. Yang, B. Yang, B. Zhang, B. Hui, et al., Qwen2.5 technical report, arXiv preprint arXiv:2412.15115 (2024).